Half-Plane Shielding: Provable Directional COLREGs Compliance for Continuous-Action Safe Reinforcement Learning

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ABSTRACT

Objectives: Autonomous surface vessels must obey the International Regulations for Preventing Collisions at Sea (COLREGs). The give-way rules are directional. A track that avoids a collision but passes on the wrong side still violates them. This paper builds a per-step guarantee of directional compliance for continuous control.
Methods: The give-way clause is written as a half-plane constraint on the plane of control commands. It is intersected with the actuator limits and a one-step collision-free condition, which gives a convex safe control set. A small quadratic program projects the command of the policy onto the nearest point of that set. The projection sits inside the environment transition, so the policy learns under its own corrected commands. The data are 2,000 simulated two-vessel encounters, split into 1,400 training and 600 test cases. Nine configurations form a single-variable ladder, each trained with eight seeds.
Findings: On every give-way step where the projection is feasible, directional compliance holds by construction, independently of the training outcome and the seed. Fallback steps are not covered. On the 600 test scenarios the median violation count is 0.53 per episode, against 1.16 for a discrete shielded baseline, at a median collision rate of 0.00%. The ablation gains are significant under a paired signed-rank test. The arrival rate is not claimed as a benefit.
Novelty: To the best of the authors' knowledge, this is the first projection-based safety mechanism that enforces directional COLREGs compliance on continuous control. The paper also states what is proved and what is not.
Practical Applications: The compliance check is independent of training. It can sit above an existing controller as a compliance layer for functional verification and liability assessment. The online cost is one small quadratic program per step, which runs in real time without special hardware.

Keywords: COLREGs compliance, safe reinforcement learning, projection-based safety shield, continuous action space, quadratic programming, unmanned surface vehicle

1 INTRODUCTION

Two vessels meet at sea. Each turns to starboard, and they pass clear of one another. What makes this work is not the judgment of either bridge team. It is a set of rules that both sides follow, and that both sides know the other will follow. When no one is on board, that predictability becomes a technical requirement. A maritime inquiry does not only ask whether contact occurred. It also asks whether the vessel acted as the rules require. An encounter that ends without contact but passes on the wrong side is still recorded as a breach.

Collision avoidance for autonomous surface vessels must therefore satisfy the International Regulations for Preventing Collisions at Sea (IMO 1972). One property of that rule set decides the shape of the problem. The give-way clauses are **directional**. In a head-on encounter both vessels turn to starboard. In a crossing encounter the give-way vessel turns to starboard and the stand-on vessel holds course and speed. In an overtaking encounter the overtaking vessel keeps clear. A control command can therefore avoid contact and still break the rules, because it turned the wrong way. Compliance cannot be verified from the outcome. It is a condition on which half-plane of the control plane each command falls in, and that is not the usual shape of a constraint in safe reinforcement learning.

Existing work follows two routes, and each gives up one property. The first route discretizes the action space and applies action masking. A rule state machine computes the compliant subset at every step and the rest is removed (Krasowski and Althoff 2024). Compliance is then guaranteed by construction, and it does not depend on the training outcome or on the seed. The price is continuous control. Quantization fixes the control resolution at the grid spacing. For this task it also has a specific cost. The rules require a give-way turn to be readily apparent, which in the control plane is a smallest admissible turn rate. A finite grid cannot land on that value. It can only take the nearest grid point at or beyond it, so every compliant turn overshoots. The second route keeps continuous control and writes the rules into the reward as a penalty (Meyer et al. 2020; Heiberg et al. 2022; Müller et al. 2026). Control resolution is preserved and the guarantee is lost, because a reward shapes expected behavior and cannot exclude any single violating command.

The intersection of continuous action spaces, provable directional compliance and COLREGs has therefore been empty. This paper fills it with a projection-based safety shield. A rule state machine returns the half-plane of compliant directions. That half-plane is intersected with the action box and with a one-step collision-free condition, which gives a convex safe control set. A quadratic program projects the continuous command of the policy onto the nearest point of that set. The step that makes this possible is a change of premise. Action masking needs the safe set to be enumerable. Projection needs it only to be convex. The shield sits inside the environment transition, so the policy learns under its own corrected commands.

The second concern of this paper is the scope of what is proved. The word provable covers different scopes in the safe reinforcement learning literature, and those scopes do not always match what a reader assumes. Each guarantee below is therefore stated with its assumptions, and the cases that are not covered are stated as well.

Contributions.

1. A projection-based COLREGs shield for continuous action spaces (Section 4.2). The compliant direction enters the quadratic program as a hard constraint. Directional compliance therefore holds by construction on every give-way step where the projection is feasible. It does not depend on the training outcome or on the seed.
2. A graded safety statement (Section 4.2.1). One-step collision freedom in the far field is a strict result. Directional compliance on every feasible give-way step holds by construction. Imminent unavoidable collision is flagged by a sufficient condition that never raises a false alarm. Each statement carries its assumptions, and the uncovered cases are named.
3. An experimental design that separates the mechanisms (Section 5.1). Nine configurations form one connected single-variable ladder. Three steps of that ladder carry a companion variable that cannot be separated, and those are listed with the ladder.

2 RELATED WORK

2.1 Maritime Collision Avoidance and COLREGs Compliance

Work on rule compliance began with model-driven methods. The velocity obstacle uses the relative velocity cone to detect risk and to select an avoiding velocity. It was given COLREGs semantics (Kuwata et al. 2014) and later extended to more general encounters (Huang et al. 2019). Model predictive control scores candidate maneuvers over a finite horizon on both risk and compliance (Johansen et al. 2016). Its branching-course variant has been tested at sea (Eriksen et al. 2019) and at full scale (Kufoalor et al. 2020). An earlier protocol-based method wrote the rules as an agreement negotiated between vessels (Benjamin et al. 2006). Recent work on this route uses control barrier functions and convex optimization (Lee et al. 2025; Patil et al. 2026), mostly on standard maneuvering models (Fossen 2011). Their guarantees come from explicit constraints. The control law is designed by hand and contains no learned policy.

Deep reinforcement learning moved the policy into the data (Sarhadi et al. 2022). Most of that work discretizes the command, for example into a finite set of rudder angles for multi-vessel encounters in confined water (Shen et al. 2019), or into compliant avoidance timing and paths (Zhao and Roh 2019; Chun et al. 2021). Discretization helps optimization and fixes control resolution at the grid spacing. A second group keeps continuous control and writes COLREGs as a soft penalty that decays with bearing and range (Meyer et al. 2020), or as a tunable risk gate (Heiberg et al. 2022). Compact reinforcement learning for obstacle avoidance has also been shown on underactuated craft (Cheng and Zhang 2018). This group is usually built on proximal policy optimization (Schulman et al. 2017), and a bounded Beta distribution (Chou et al. 2017) can parameterize the continuous command so that samples are not clipped at the boundary of the action box. Control resolution is preserved. Rule compliance is induced by the reward, so no single violating command is excluded.

2.2 Safe Reinforcement Learning and Provable Compliance

Safe reinforcement learning (García and Fernández 2015) studies how to attach hard guarantees to a learned policy. Writing safety as a constrained Markov decision process and solving it with a Lagrangian method is a common route. What it guarantees is constraint satisfaction in expectation, which again excludes no single command. Action masking removes unsafe commands from a discrete action space according to a temporal logic specification (Alshiekh et al. 2018). A realizable form of it has been extended to continuous action spaces for general tasks, without maritime rules (Kim et al. 2024). In continuous control, quadratic programs built on control barrier functions (Ames et al. 2019) and predictive safety shields (Wabersich and Zeilinger 2021) give per-step hard guarantees. The latter has been attached to maritime reinforcement learning, where it constrains collision only and carries no directional clause (Vaaler et al. 2024). The closest mechanism is a safety shield that linearizes the constraint at the current state. It then projects the command of the policy onto the resulting half-space in closed form (Dalal et al. 2018). The projection used here has the same form. What is added is the encoding of the directional give-way clause as a half-plane in the control plane. The enforced object is then rule compliance rather than a state bound.

Table 1 places this work on four dimensions and shows where the gap is. A hard guarantee of rule compliance in a maritime setting had been demonstrated only on a discrete action space (Krasowski and Althoff 2024). That work provides the scenario library and the state machine thresholds used here. Action projection under continuous control is listed there as future work (Kochdumper et al. 2023). Later work on that direction moved to continuous control but approximated compliance with a soft reward, without a hard guarantee (Müller et al. 2026). The present work gives a constructive guarantee of directional compliance under continuous control. The reachable set is represented by a convex over-approximation and the correction by a quadratic program (Markgraf et al. 2026), rather than by a higher-order set representation, which keeps the per-step cost small.

A directional constraint tightens the feasible set in close quarters and can cost terminal efficiency, so potential-based shaping is used to recover it. Potential-based shaping does not change the set of optimal policies (Ng et al. 1999). A naive distance potential tends to stall near the goal (Trott et al. 2019), and a single continuous potential does not remove that behavior (Müller and Kudenko 2025). A linear kernel with arrival as a terminal condition is used instead (De Lellis et al. 2024). The effect of a shield on performance has been studied separately. Including the filter during training and penalizing the correction improves sample efficiency (Pizarro Bejarano et al. 2025), a sufficiently permissive filter does not cost asymptotic optimality (Oh et al. 2025), and a safety term that opposes the goal term causes stalling short of the goal (Grover et al. 2023).

3 PROBLEM FORMULATION

3.1 Vessel Kinematics

The controlled vessel is modeled as a point mass with a yaw constraint. Its state is $s = [p_x, p_y, \theta, v]^\top \in \mathbb{R}^4$, which collects position, heading, and speed. Its controls are the longitudinal acceleration a and the turn rate ω,

$$\dot{s} = \begin{bmatrix} v\cos\theta, & v\sin\theta, & \omega, & a \end{bmatrix}^\top, \qquad u = [a, \omega]^\top. \tag{1}$$

The controls obey $|a| \le a_{\max}$ and $|\omega| \le \omega_{\max}$, and the speed obeys $0 \le v \le v_{\max}$. A command is held constant over one decision period Δt. Headings and bearings are wrapped to $(-\pi, \pi]$. A relative bearing is positive to starboard.

TABLE 1 Position of related work. A per-step hard guarantee means the class of commands the method can exclude at *every* decision step, not constraint satisfaction in expectation.

Representative work	Action space	Compliance is imposed by	Per-step guarantee	Learned
Model-driven, no learning				
(Kuwata et al. 2014)	Continuous	Velocity obstacle geometry	Geometric	-
(Johansen et al. 2016)	Branches	Cost ranking	-	-
(Lee et al. 2025; Patil et al. 2026)	Continuous	Barrier function	State bound	-
Learning, compliance from a soft reward				
(Meyer et al. 2020)	Continuous	Soft reward	-	✓
(Heiberg et al. 2022)	Continuous	Soft reward, risk gate	-	✓
(Müller et al. 2026)	Continuous	Soft reward, falsification	-	✓
Learning, compliance from a hard constraint				
(Alshiekh et al. 2018)	Discrete	Action masking	State bound	✓
(Dalal et al. 2018)	Continuous	Linearize, then project	State bound	✓
(Vaaler et al. 2024)	Continuous	Predictive safety shield	Collision only	✓
(Krasowski and Althoff 2024)	Discrete (49)	Action masking	**Direction**	✓
Ours	**Continuous**	**Projection (QP)**	**Direction**	✓

The turn rate follows the usual mathematical convention and is positive counter-clockwise, so a turn to starboard has $\omega < 0$.

The speed bound is applied after the step is integrated, so the speed can briefly exceed $v_{\max}$ within a step. Every one-step displacement bound in this paper therefore uses $v_{\max}\Delta t + \frac{1}{2}a_{\max}\Delta t^2$, and the speed at the end of a step is at most $v_{\text{bnd}} = v_{\max} + a_{\max}\Delta t$.

Two reasons fix this level of detail. The give-way rules govern changes of course and of speed, and Equation (1) takes both as control commands. The rules can therefore be written as constraints on u with nothing in between, and that is what lets the whole safety mechanism live in the action space. A finer hydrodynamic model would add coefficients that differ by vessel and are hard to identify, and the rule constraints would then be entangled with vessel parameters. Wind, current, and waves are not modeled.

Vessel and occupancy. The subject is a large container vessel of length 175 m and beam 25.4 m. Its limits are $a_{\max} = 0.24$ m/s^2, $\omega_{\max} = 0.03$ rad/s, and $v_{\max} = 9.5$ m/s. The decision period is 10 s, and an episode lasts at most 170 steps, which is 1,700 s. The circumscribed radius of the hull is 88.4 m and the inscribed radius is 12.7 m.

The turn rate limit is about 1.7° per second. A readily apparent turn of 20° therefore needs more than ten seconds, which is longer than one decision period. The vessel maneuvers more slowly than it decides. A safety constraint must look ahead for that reason alone.

The hull is represented by a rectangle, and two circles approximate it for two different purposes. The circumscribed circle over-approximates the hull, and it supports conditions of the form “contact is impossible”. The inscribed circle under-approximates the hull, and it supports conditions of the form “contact is certain”. One circle in each direction is what keeps both kinds of conclusion one-sided. The target vessel is represented by a rectangle as well, and it is extrapolated at constant velocity whenever a prediction is needed.

3.2 Observation and Action Spaces

The observation is a real vector of 27 entries in four groups. Four entries give the state of the own vessel. Six give the relation to the goal. Twelve give the traffic situation. Four bearing sectors are used, and each contributes the nearest target vessel in it. Each such vessel gives its distance, its relative bearing, and the change of that distance over one step. Five are termination flags. The observation layer outputs raw physical quantities, and normalization is done in the training wrapper. Organizing traffic by sector keeps the dimension independent of the number of target vessels. That solves the dimension only. The shield deployed here is implemented for a single target vessel, and it raises an error rather than running in a degraded mode when more are present.

Two action boxes. The action is the two-dimensional command $u = (a, \omega)$, and two boxes have to be distinguished. The policy acts in the narrower operating box, whose half-widths are 0.048 m/s^2 and 0.018 rad/s. These are exactly the span of the 7×7 grid used by the discrete configurations, so the two action spaces carry the same control authority and differ only in resolution. The shield and the emergency controller act in the wider box of the actuator limits, $U_{\text{box}} = \{|a| \leq a_{\max}\} \cap \{|\omega| \leq \omega_{\max}\}$.

The widening is necessary. A shield confined to the narrow box would declare a step infeasible whenever that box and the collision-free constraint disagree, even though the step is in fact solvable. The cost is that the applied command can fall outside the policy action box. That fact is returned in the observation, and saturation is counted against each box separately.

3.3 Compliance as a Per-Step Constraint

We consider a two-vessel encounter in open water under five assumptions. (A1) Open water, with no traffic separation scheme, no static obstacle, and no shoreline constraint. (A2) One controlled vessel and one target vessel, both power-driven. (A3) The dynamics of the controlled vessel are Equation (1), applied with zero-order hold at period Δt. (A4) The current state of the target vessel is available without measurement error. (A5) No give-way obligation is already in force at the initial instant.

Assumption (A2) is not a simplification adopted for convenience. It is the scope of the convention itself. The rules are written for a pair of vessels (IMO 1972). Three or more vessels can meet at once, and their obligations can then conflict. One case is an obligation to hold course for one vessel and to give way to another. The convention does not say which obligation prevails.

Let ρ_t be the encounter situation at time t. Let $U_{\text{colregs}}(\rho_t) \subseteq \mathbb{R}^2$ be the set of commands that the rules allow in that situation, and let $U_{\text{cf}}(s_t)$ be the collision-free set. The problem is to find a policy

$$\pi : S \to \mathbb{R}^2 \quad \text{such that} \quad u_t \in U_{\text{colregs}}(\rho_t) \cap U_{\text{cf}}(s_t) \quad \text{for all } t. \tag{2}$$

Equation (2) is not the same as writing compliance into the reward. It constrains every command that is executed. It does not ask for an expected value to be optimal.

When the action set is finite, Equation (2) has a direct solution. Every action is enumerated, tested, and masked out if it does not comply. That route needs the actions to be enumerable, so it is not compatible with continuous control. Rudder and propulsion are continuous quantities, and quantizing them costs both smoothness and resolution. This paper replaces enumeration by projection. The give-way clause is a half-plane constraint on the command (Section 4.1). The collision-free condition can be written as a linear inequality (Section 4.2). The feasible set of Equation (2) is therefore the intersection of a rectangle with a few half-planes in the plane, which is a convex polygon. The projection onto a convex set exists and is unique, and it is obtained from a very small quadratic program.

4 METHODOLOGY

A note on the guarantees. Three propositions are stated in this section. Each one is placed immediately after the mechanism it applies to, rather than collected in a separate block. Each is given as a statement, its assumptions, a proof sketch, and its scope. The scope paragraph names the cases that the proposition does not cover.

4.1 COLREGs Obligations as Half-Plane Constraints

The rule state machine sorts the current encounter into six situations,

$$\rho \in \{\rho_0 \text{ conflict-free},\ \rho_1 \text{ stand-on},\ \rho_2 \text{ head-on},\ \rho_3 \text{ crossing},\ \rho_4 \text{ overtaking},\ \rho_5 \text{ emergency}\},$$

and the decision is a conjunction of three groups of conditions. Each condition is a predicate, that is, a named test on the two states that returns true or false.

The first group is the bearing sector, taken along the sidelight boundaries of the convention. The second group is the heading relation, which separates nearly anti-parallel from nearly parallel headings. The third group is collision possibility. It has two parts. The speed set of the own vessel must intersect the collision cone of the target vessel. The relative speed must also close the current center distance in time. The forward sector has a half-angle of 5° and the side sectors end at 112.5°. The overtaking band has a half-angle of 67.5°. The occupancy of the target vessel is inflated by three times its length, and the check horizon is 420 s.

Four situations carry an obligation. More than one of their predicates can hold at the same instant, so the four are tested in a fixed priority order and the first match wins. The rows below are written in that priority order, which is not the order of the indices,

$$
\begin{aligned}
\rho_2 \text{ head-on}: &\quad \mathrm{CP} \wedge \mathrm{Fr}(s_e, s_m) \wedge \mathrm{Ap}, \\
\rho_3 \text{ crossing}: &\quad \mathrm{CP} \wedge \mathrm{Ri}(s_e, s_m) \wedge \mathrm{Tl}, \\
\rho_4 \text{ overtaking}: &\quad \mathrm{CP} \wedge \mathrm{Be}(s_m, s_e) \wedge (v_e > v_m) \wedge \mathrm{Sd}, \\
\rho_1 \text{ stand-on}: &\quad \big[\mathrm{CP} \wedge \mathrm{Le}(s_e, s_m) \wedge \mathrm{Tr}\big] \ \vee\ \rho_4(s_m, s_e),
\end{aligned}
\tag{3}
$$

and all other cases fall into ρ_0. Stand-on is tested last, and its definition reuses the overtaking predicate with the two vessels exchanged. Overtaking and being overtaken swap the two vessel arguments, because overtaking places the own vessel in the aft sector of the target vessel. The indices ρ_0 to ρ_5 are identifiers rather than a ranking.

Entry into a give-way situation has two parts. A give-way predicate must not hold now. It must hold at every decision step of a 60 s reaction window under constant-velocity extrapolation. The first part separates "about to hold" from "already holds", and it creates one asymmetry that Section 4.3.3 treats. The emergency situation is entered by a stronger set-based prediction with a horizon of 180 s.

Compliant control set. Each situation maps to an explicit constraint set on the command,

$$
U_{\mathrm{colregs}}(\rho) = \begin{cases}
U_{\mathrm{box}}, & \rho = \rho_0, \\
\{|a| \le \varepsilon_a\} \cap \{|\omega| \le \varepsilon_\omega\}, & \rho = \rho_1, \\
\{\omega \le -\omega_{\mathrm{turn}}\}, & \rho \in \{\rho_2, \rho_3\}, \\
\{\omega \le -\omega_{\mathrm{turn}}\} \text{ or } \{\omega \ge +\omega_{\mathrm{turn}}\}, & \rho = \rho_4, \\
U_{\mathrm{box}}, & \rho = \rho_5.
\end{cases}
\tag{4}
$$

Head-on and crossing require a turn to starboard. Overtaking allows a turn to either side. Stand-on holds course and speed. Each set is a half-plane or an interval on the plane of control commands. A set of this shape can be entered by projection, so the commands never have to be enumerated.

Quantifying a readily apparent action. The convention asks a give-way action to be large enough to be seen in time. We quantify this as a course change of at least a given threshold within the maneuver window. With a window of 40 s and a threshold of 20°,

$$
\omega_{\mathrm{turn}} = \frac{\Delta_{\mathrm{lt}}}{T_M} = \frac{20^\circ}{40\ \mathrm{s}} \approx 0.008727\ \mathrm{rad/s}. \tag{5}
$$

This is the smallest compliant turn rate. Any turn at least this large complies, and the bound cannot be lowered.

The predicate structure and the numerical thresholds follow the temporal-logic formalization of the maritime rules (Krasowski and Althoff 2024). Three details are settled here. The inflation radius and the half-width of the speed set are stated explicitly. Entry into a give-way situation is made symmetric (Section 4.3.3). The predicate set is then mapped to a constraint set on continuous commands.

4.2 Safe Control Set and Projection

4.2.1 One-Step Collision-Free Constraint

At each decision step the applied command must keep the two hulls apart at the end of the next step. Written directly, that condition is not convex. We replace it by a convex subset of itself, built in three steps.

First, the occupancy of the target vessel after one step is over-approximated by a disk. The center follows a constant-velocity extrapolation, and the radius includes a safety margin. The own vessel is over-approximated by its circumscribed circle. The two hulls then stay apart whenever the center distance is at least the sum of the two radii, written d_{safe}.

Second, let n be the unit vector from the target vessel to the nominal end point of the own vessel, and let δ be that distance. The end point follows from the policy command clipped to the action box.

Third, the end point is expanded to first order in the command, $p_e(u) \approx p_{\mathrm{nom}} + J(u - u_{\mathrm{nom}})$. The end point must lie on the safe side of the separating hyperplane. This gives one linear inequality in the command:

$$
\underbrace{-n^\top J}_{g^\top}\, u \ \le\ \underbrace{-\big(d_{\mathrm{safe}} - \delta + n^\top J\, u_{\mathrm{nom}}\big)}_{h}, \qquad U_{\mathrm{cf}}(s) = \{u : g^\top u \le h\}. \tag{6}
$$

A first-order expansion is neither an inner nor an outer approximation. The sign of the remainder changes with the curvature. Equation (6) alone therefore does not establish conservatism. The margin inside the safety distance is what does. With that margin the constraint acts at a center distance of 590–840 m, with a median of 714 m over this library. Hull contact needs a much smaller distance. The margin absorbs the linearization residual. The cost is a tighter constraint that rejects some commands which are in fact safe.

The linearization holds only locally. In the far field a sufficient condition is available that does not rely on it.

Proposition 1 (One-step collision freedom in the far field)**.** *Let d be the center distance between the own vessel and a single target vessel. If*

$$d \ \geq \ d_{\text{safe}} + v_{\text{obs,max}} \Delta t + \rho_{\text{ego}}, \tag{7}$$

then the two conservative circumscribed occupancies do not intersect after one decision step. Here d_{safe} is the sum of the two circumscribed radii, and ρ_{ego} is the largest displacement of the own vessel in one step.

Assumptions. A single target vessel. The speed of the target vessel is at most 9.5 m/s. That bound is not implied by the dynamics and is stated separately; the largest value measured over the 2,000 scenarios of the library is 7.10 m/s. The target vessel holds its speed within the step.
Proof sketch. By the triangle inequality, the center distance after the step is at least d minus the one-step displacement bound of each vessel. It is therefore at least d_{safe}.
Scope. The triangle inequality does not depend on direction, so a head-on encounter is the worst case. With the constants of this library the threshold is about 764 m. The statement covers one step. It does not extend to the steps after it, and it does not cover the near field.

4.2.2 Projection as a Quadratic Program

The safe control set is the intersection of three sets,

$$U_{\text{safe}}(s, \rho) = U_{\text{box}} \cap U_{\text{colregs}}(\rho) \cap U_{\text{cf}}(s), \tag{8}$$

and it is a convex polygon in $\mathbb{R}^2$. The applied command is the Euclidean projection of the policy command onto that set,

$$u_{\text{safe}} = \Pi_{U_{\text{safe}}}(u_{\text{policy}}) = \arg \min_{u \in U_{\text{safe}}} \|u - u_{\text{policy}}\|_2^2. \tag{9}$$

The feasible set is an intersection of half-planes, so it can be written as $U_{\text{safe}} = \{u : Au \leq b\}$. The first four rows are the action box, the fifth is the compliant-direction half-plane, and the sixth is the collision-free constraint of Equation (6). The program has two variables and at most six constraints. It is small enough to solve online at every decision step.

Proposition 2 (Directional compliance on every give-way step)**.** *Consider any decision step that the rule state machine assigns to a give-way situation and on which the projection is feasible. The applied command u_{safe} then lies in the compliant-direction half-plane.*

Assumptions. A single target vessel, and a correct decision by the state machine. The second assumption inherits one conservative bias of the collision-possibility predicate. That predicate uses the center distance, and it can miss a trailing encounter in which the two speeds are close.
Proof sketch. The compliant half-plane enters the quadratic program as a hard constraint. Any feasible point satisfies it, and the projection returns a feasible point. The statement holds by construction. It does not depend on the training outcome or on the random seed.
Scope. Emergency situations are not covered, because their branch bypasses the compliance constraint by design. Steps on which the safe control set is empty and the fallback is used are not covered either.

4.2.3 Fallback and the Unavoidable-Collision Criterion

When the safe control set is empty, the projection has no solution. A substitute rule then supplies the command. We call it the fallback, and it carries no guarantee. The fallback branches on the encounter situation. It does not try a fixed list of options in order. The two branches below are mutually exclusive.
Emergency branch. An emergency situation takes a separate path. No collision-free constraint is built, and Equation (9) is not solved. A stateful geometric controller supplies the command instead, and its lifetime is one emergency

event. It has three modes. A near head-on approach ahead is handled by tracking a fixed lateral point fixed at the onset of the emergency. A target vessel astern that can be cleared by acceleration is handled by a straight-line escape inside the reaction window. Every other case tracks a point that moves with the target vessel. One position-tracking law converts all three into a control command. This controller offers no guarantee. Its purpose is a defined behavior once no compliant feasible command remains.

Other situations. Outside an emergency situation the fallback is used only when the safe control set is empty. A degenerate case is tested first. If the linearization gives no usable separation direction, the command is projected onto the action box alone. Otherwise the direction constraint is relaxed, and the collision risk is minimized if that also fails. The last two branches solve on the actuator limits and drop the compliance constraint as a whole. The turn rate can then leave the policy action box, so saturation is reported against each box separately.

These branches are needed in the implementation, and on the scenario library used here they almost never fire. If the own vessel holds course and speed, the median closest center distance to the target vessel is about 1,300 m. The safe control set is therefore rarely empty. Purpose-built conflict cases are constructed so that the branches are exercised, and their outcome is reported with the shield measurements.

The emergency branch offers no guarantee. It can still decide whether the situation was already beyond recovery.

Proposition 3 (Conservative criterion for an unavoidable collision)**.** *Consider a single target vessel at constant velocity. Let $R_{\text{box}}(t)$ be an over-approximation of the reachable centers of the own vessel, and let $O(t)$ be the hull rectangle of the target vessel. If*

$$\exists t^* \in [0,T]: \quad R_{\text{box}}(t^*) \subseteq \big(O(t^*) \oplus \text{disk}(r_{\text{insc}})\big), \tag{10}$$

then the collision is unavoidable, and every feasible control sequence meets that target vessel at t^*.

Assumptions. A single target vessel at constant velocity. The hull size of the target vessel must not be overstated, since a beam larger than the true value makes the test unreliable. The horizon is 120 s, evaluated at 241 equally spaced instants.

Proof sketch. The rate of change of the velocity vector is bounded. A double integration gives the two semi-axes of R_{box}, which therefore over-approximates the true reachable centers. The inscribed circle under-approximates the hull. Once every reachable center lies inside the inflated occupancy of the target vessel, no control command can avoid contact.

Scope. The criterion is one-sided. It can miss a case, and it never raises a false alarm. A negative result therefore does not mean that the collision is avoidable. The criterion also acts late rather than early, and maneuvering target vessels are not covered. On 20,000 random scenarios and 735 selected scenarios it produced no false positive. After the envelope was hardened, 1,500 further fuzz cases produced none either.

4.3 Shielded Policy Optimization

The shield is not applied after the policy as a post hoc correction. The projection of Equation (9) is placed inside the environment transition. The policy emits u_{policy}, the environment applies u_{safe}, and it returns the successor state and the reward. The policy therefore faces the composed transition

$$s_{t+1} \sim P\big(\cdot \mid s_t,\, \Pi_{U_{\text{safe}}(s_t,\rho_t)}(u_t)\big), \qquad u_t \sim \pi(\cdot \mid s_t), \tag{11}$$

rather than the $P(\cdot \mid s_t, u_t)$ of a post hoc filter. The policy learns on the transition it will actually meet, so it adapts to the projection during training instead of meeting an unfamiliar filter afterwards.

Learning algorithm. All configurations are trained with proximal policy optimization (Schulman et al. 2017). The continuous-action configurations use its standard form. The discrete-action baselines use its maskable variant, because action masking is what makes a discrete shield possible. Both action spaces therefore share one policy-gradient algorithm, which removes a source of difference between them. The masking itself remains a difference, and it is intrinsic to the discrete method rather than a choice made here.

4.3.1 Reward Function

The reward is a sum of five terms,

$$r = r_{\text{sparse}} + r_{\text{goal}} + r_{\text{velocity}} + r_{\text{deviate}} + r_{\text{colregs}}. \tag{12}$$

A sparse terminal term scores arrival, timeout, stalling, and collision, and it charges a fixed penalty for every step that enters emergency control. A dense approach term rewards the change in distance to the goal over one step. A speed term penalizes speed outside the cruise band [2.5, 8.0] m/s. A cross-track term penalizes lateral deviation from the start-to-goal line and saturates at 2,000 m. A soft compliance term decays with distance. It is active only in the reward-shaping baseline and in the discrete configurations. Every continuous configuration here sets it to zero, because compliance is carried by the hard constraint of Equation (4).

The five terms are taken from (Krasowski and Althoff 2024), with three deviations that we declare. The approach term uses the difference of distances between two steps. The soft compliance term is rescaled, because its original calibration is set at a scale about fifty times smaller than the scenarios used here. One clause of the emergency-release test is read by its physical meaning rather than by its literal sign, since the two disagree.

4.3.2 Bounded Action Distribution

A continuous policy usually emits an unbounded Gaussian and clips the sample to the action box before it reaches the environment. That combination conflicts with Equation (9), for two reasons that both come from the clipping. First, common implementations feed the clipped action to the environment but the unclipped action to the policy gradient. Once the mean leaves the box, moving further out changes no reward, the gradient reaches a plateau, and any penalty on action smoothness loses its grip. Second, the differential entropy of a Gaussian has no upper bound, while the entropy coefficient is a constant. The standard deviation therefore has a one-way channel to grow. Its growth pushes still more samples outside the box.

We use a Beta distribution with support on a closed interval. Actions can then never leave the box, clipping becomes the identity map, both channels close, the smoothness penalty is differentiable everywhere, and the entropy is bounded above. The network emits two reals per control axis, and

$$\alpha = \mathrm{softplus}(\tilde{\alpha}) + 1, \quad \beta = \mathrm{softplus}(\tilde{\beta}) + 1, \qquad u = u_{\mathrm{lo}} + (u_{\mathrm{hi}} - u_{\mathrm{lo}})z, \quad z \sim \mathrm{Beta}(\alpha, \beta) \tag{13}$$

gives the shape parameters and the applied action. Evaluation uses the mode instead of a sample. The offset of one keeps both shape parameters above unity, and that constraint cannot be dropped. Below unity the Beta density is U-shaped and diverges at the two ends, and the mode then sits at an endpoint. Because evaluation always takes the mode, such a policy would silently corrupt every reported number, and the aggregate metrics would give no sign of it.

4.3.3 Symmetric Entry into Give-Way Situations

In the state machine, entry into a give-way situation is not symmetric. From the conflict-free situation it is triggered only by the persistence condition. From the stand-on situation a separate immediate condition also applies. We add the symmetric branch. When the persistence condition does not fire, the immediate condition is checked, and the give-way situation is entered if it already holds. The obligation is then recognized at the same moment on both routes. The change alters the command that the shield applies, so training and evaluation must use the same setting.

A purpose-built case shows why the branch is needed. Suppose an encounter already satisfies a give-way predicate at the initial instant. The clause "does not hold now" can then never be satisfied. The persistence condition never fires, and the encounter is never recognized as a give-way situation. On a set of such adversarial cases the persistence route alone gives 0 give-way decision steps. With the immediate branch it gives 201. These cases are synthetic and illustrate the mechanism, so they are reported apart from the main results.

4.3.4 Training Setup

Apart from the entropy coefficient, which is set to 0.01, and the bounded action distribution, every hyperparameter takes the default of the implementation library and is left untuned. The libraries are Stable-Baselines3 2.3.2 (Raffin et al. 2021) and sb3-contrib 2.3.0, and the advantage is computed by generalized advantage estimation (Schulman et al. 2016). The discrete and continuous configurations agree on every one of these values, so the hyperparameters are not a source of difference between them. Running normalization of the observation and the reward is not optional. Without it the large-scale approach term in Equation (12) drives the policy to a degenerate solution that holds station.

5 EXPERIMENTS

5.1 Scenarios and Configurations

The experiments use the two-vessel encounter library of the CommonOcean benchmark (Krasowski and Althoff 2022), which holds 2,000 scenarios. Each scenario fixes the initial pose and speed of the controlled vessel. It also fixes the initial state and the planned track of the target vessel. It ends with a goal region that has a position gate, a heading gate, and a time limit. The encounter geometry is generated programmatically over a range of relative bearings, crossing angles, and closing speeds. Target vessels are 200 to 300 m long, and an episode lasts at most 170 decision steps.

Classifying the library by the geometric predicates returns crossing and head-on only. **The library contains no overtaking scenario**, and the official test set holds 395 crossing and 205 head-on cases. The overtaking branch of Equation (4) therefore cannot be tested empirically here. The conclusions state this.

The official split of 1,400 training and 600 test scenarios is used. The training side is split again into 1,300 for training and 100 for validation. The validation set is used only for monitoring during training and for selecting a checkpoint. The test set is never accessed during training or tuning. An independent recount confirmed that the two sets are disjoint, so the evaluation sample size is 600.

Table 2 lists nine configurations. They form one connected ladder, and each level changes a single variable from the level before it. The contribution of each mechanism can therefore be read on its own.

TABLE 2 The nine configurations

Configuration	Cont.	Shield	Masking	Soft rew.	Shaping	Bounded	Sym. entry
Discrete baseline	×	×	×	×	×	-	-
Discrete, soft reward	×	×	×	✓	×	-	-
Discrete, masking	×	×	✓	✓	×	-	×
Continuous, no shield	✓	×	×	×	×	×	×
Continuous, shield	✓	✓	×	×	×	×	×
Ablation reference	✓	✓	×	×	✓	×	×
Ablation, bounded only	✓	✓	×	×	✓	✓	×
Ablation, symmetric only	✓	✓	×	×	✓	×	✓
Ours	✓	✓	×	×	✓	✓	✓

Notes. A check mark is on, a cross is off, and a dash is not applicable. Rows run from the discrete baseline to Ours. Three steps of the ladder carry a companion variable that cannot be separated. The action space changes together with the policy distribution, the continuous-only shaping is a bundle of four items, and the bounded distribution also changes the initial exploration scale. Those three steps can only be attributed as a whole.

What the unshielded continuous configuration can show. A continuous-action configuration without a shield can only take the plainest form, namely an unbounded Gaussian, an asymmetric give-way entry, and no shaping. Giving it the continuous-only shaping would break the symmetry with the discrete baseline, and the attribution would fail. The unbounded Gaussian is also the form that pushes many samples past the action box. A small re-evaluation was run during the exploration phase, and that setting is not the one used in the formal experiment. About 73% of the decision steps of that policy sat against the boundary of the action box. The gain attributed to the shield is therefore measured against the weakest form of continuous action. For the same reason the step from the discrete baseline to that configuration can be negative, and it should be read as a conservative lower bound.

5.2 Metrics and Evaluation Protocol

Metrics. The main metrics are the arrival rate, the collision rate, the COLREGs violations per episode, the share of steps under emergency control, and the episode duration. The violation count is split into give-way and stand-on violations. Control quality, safety margin, shield behavior, and the single-step solve time are also reported. Every metric is broken down by encounter situation as well. The reward function differs across configurations, so the episode return is never compared across them.

How violations are counted. Violations do not come from the situation decision of the shield. They come from an offline scorer that is independent of the shield. The scorer sees only the trajectory. At each step it re-decides, from the bare situation predicates, whether the own vessel is under a stand-on or a give-way obligation. A stand-on violation is counted per step, once the signed heading change accumulated over the obligation period exceeds the stand-on tolerance, and the accumulator is reset when the obligation ends. A give-way violation is counted per encounter, and an encounter counts as a violation only when no turn in the required direction ever reaches the readily apparent threshold. The scorer applies one rule to this method and to every baseline, including the geometric controllers that have no state machine, so the comparison is made under one condition.

The counting window and the shield do not coincide. The scorer accumulates the heading change over a whole obligation period, while the shield constrains the change inside a single maneuver window. The two windows have different lengths. Emergency situations and steps that fall back also bypass the compliance constraint by design. The shield therefore does not guarantee a violation count of zero, and this paper does not claim one. In particular, Proposition 2 guarantees the turn direction on every give-way step. It does **not** imply a give-way violation count of zero. How often the mismatch occurs was not measured.

Training budget. Each configuration is trained with 8 random seeds, one run per seed. The 72 runs together cover about 730 million environment steps. The scenario split is fixed by a separate seed, so every run sees the same scenarios. Every configuration inside one table is evaluated on one machine in one pass, so that machine differences do not enter the evaluation. Machine time did not allow the same during training. For 5 of the 8 seeds all configurations were trained on one machine, and the other 3 seeds are spread over two or three machines. Those machines run the same image and the same training code, with a byte-identical fingerprint of the training path, and with the thread count locked to one. **No bit-level cross-machine check was run on this batch, so machine differences are not excluded. This is listed as a limitation.**

Checkpoint selection. Twenty checkpoints are saved per run at even intervals, and each is evaluated once on the validation set. The selection rule was fixed in advance and has no tunable part. The candidates are those twenty checkpoints, the criterion is the highest arrival rate on the validation set, ties go to the earlier checkpoint, and the test set is never touched. **The rule is itself biased, so two versions are reported.** The validation set holds only 100 scenarios, and the binomial standard error of the arrival rate is about 4 points. Taking the maximum over 20 candidates lifts the reported value by about 7.5 points in expectation. That is four times the drift of about 1.6 points that the same-machine rule was written to prevent. The best-checkpoint version and the last-checkpoint version are therefore both reported.

Reading the figures. Figures 1 and 2 are computed on the same test set as Table 3. Figures 3 and 4 are recorded during training and are labelled as such. Violation counts in the figures are the sum of stand-on and give-way violations.

Statistics. The paired test is a per-seed paired sign test. With 8 seeds the smallest two-sided value it can reach is $p = 2/2^8 = 0.0078$. Error bars are bootstrap 95% intervals resampled over seeds. A run counts as converged when its arrival rate on the validation set reaches 50%. Diverged seeds are counted in the main table as they are, and a converged-seeds-only version is also reported.

5.3 Main Comparison

TABLE 3 Main comparison, all seeds. Official test set, $N = 600$, at the checkpoint selected on the validation set.

Configuration	n	Conv.	Arrival % [CI]	Coll. %	Viol./ep.	Turn Δ	Thr. Δ	Emerg. %
Discrete baseline	8	**8/8**	97.8 [97, 98]	2.00	3.04	0.0149	0.0160	0.00
Discrete, soft reward	8	7/8	**98.2 [97, 98]**	1.58	2.95	0.0141	0.0169	0.00
Discrete, masking	8	6/8	93.3 [10, 96]	0.08	1.16	0.0112	0.0205	4.64
Continuous, no shield	8	7/8	98.0 [96, 99]	1.25	4.44	0.0267	0.0106	0.00
Continuous, shield	8	5/8	69.0 [16, 91]	0.17	1.02	0.0246	0.0158	2.86
Ablation reference	8	5/8	84.5 [9, 90]	0.33	1.13	0.0140	0.0059	3.36
Ablation, bounded only	8	7/8	90.8 [66, 92]	0.17	0.67	0.0049	0.0076	2.27
Ablation, symmetric only	8	6/8	75.5 [4, 89]	**0.00**	0.82	0.0167	0.0053	2.61
Ours	8	**8/8**	88.2 [84, 90]	**0.00**	**0.53**	0.0054	0.0078	2.42

Notes. The best value in each column is in bold. The turn and throttle increment columns are **not** marked, because they cannot be compared across action spaces (Section 4.3.4). The share of emergency steps is descriptive and is neither good nor bad.

Three readings follow from Table 3. First, Ours has the lowest violation count per episode of all nine configurations, at 0.53, against 1.16 for the discrete safe baseline and 4.44 for the unshielded continuous configuration. A per-seed paired sign test gives 8 wins out of 8 against every other configuration except the two shielded ablations, at $p = 0.0078$. Second, Ours is the only shielded configuration that reaches the convergence criterion on all 8 seeds.

Third, the arrival rate of Ours is 88.2%, which is below the three unshielded configurations and below the discrete safe baseline. That gap is the cost of the shield and it is not claimed as a benefit anywhere in this paper. Table 3 is also reported in a converged-seeds-only version, because a diverged seed pollutes the per-episode averages and the direction of that pollution differs by configuration. Per-seed raw values accompany the table.

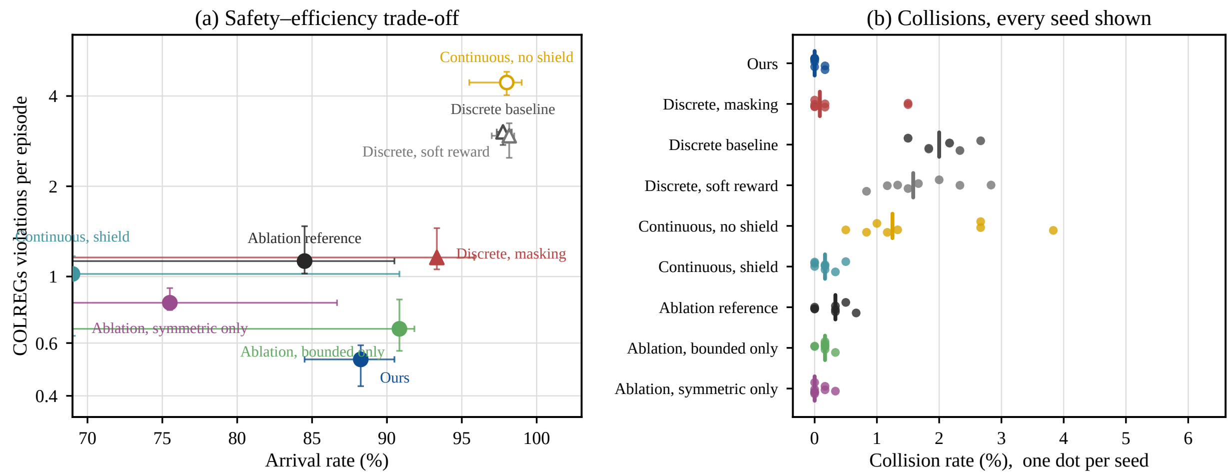

Figure 1 Safety and efficiency. Official test set, the same source as Table 3. Points are medians across seeds and error bars are bootstrap 95% intervals resampled over seeds. Circles are continuous action spaces, triangles are discrete ones, and filled markers carry a shield. The lower right of panel (a) is the ideal region. The bars are wide because several configurations contain a diverged seed.

5.4 Ablation Study

The four ablation configurations form a complete 2×2 design over the bounded action distribution and the give-way entry. Everything else is identical, and the comparison is paired by seed. The bounded distribution lowers the median turn increment by 65%, with all 8 seeds moving in the same direction and a paired signed-rank value of $p = 0.008$. It lowers the violations per episode by 41%, again 8 of 8, with $p = 0.008$. The symmetric entry alone **raises** the turn increment by 19%, with $p = 0.016$, and it lowers the violations per episode by 27%, with $p = 0.008$. With both changes on, the turn increment falls by 61% and the violations by 53%. Neither can be separated statistically from the bounded distribution alone, at $p = 0.74$ and $p = 0.11$. The two changes therefore divide the work. Smoothness comes entirely from the bounded action distribution, and both contribute to compliance.

This group is the only place where the smoothness result can be read. All four configurations run the same action-increment penalty, so the difference can be attributed to the bounded action distribution rather than to an extra reward term. A comparison against the discrete baseline would not be valid here, because that penalty applies to a quantity the discrete configurations do not have. Adding the bounded distribution also carries a companion variable, since the initial exploration scale differs by about a factor of 2.5 on the acceleration axis. A converged-seeds-only version is given for this group as well.

5.5 Training Reliability and Sample Efficiency

Figure 3 is taken from training. It answers two questions that do not need the re-evaluation, namely whether the shield costs learning speed, and how many seeds of each configuration reach a working policy.

First, there is a cost, and it appears in learning speed. On the training scenarios the unshielded discrete baseline reaches an arrival rate of 96% by 2 million steps. The shielded continuous configuration is at 33% at that point, and it reaches 88% at the end of training. The shielded discrete baseline lies between the two, at 51% and 93% (Figure 3b). Panels (c) and (d) give what is bought. The unshielded configurations hold a collision rate near 1.8% throughout training and more than 2.4 violations per episode. The shielded configurations sit at a collision rate of zero and near 0.5 violations per episode. The shield removes risk from the outcome, and it narrows the exploration space at the same time. Slower learning is the direct consequence of that narrowing.

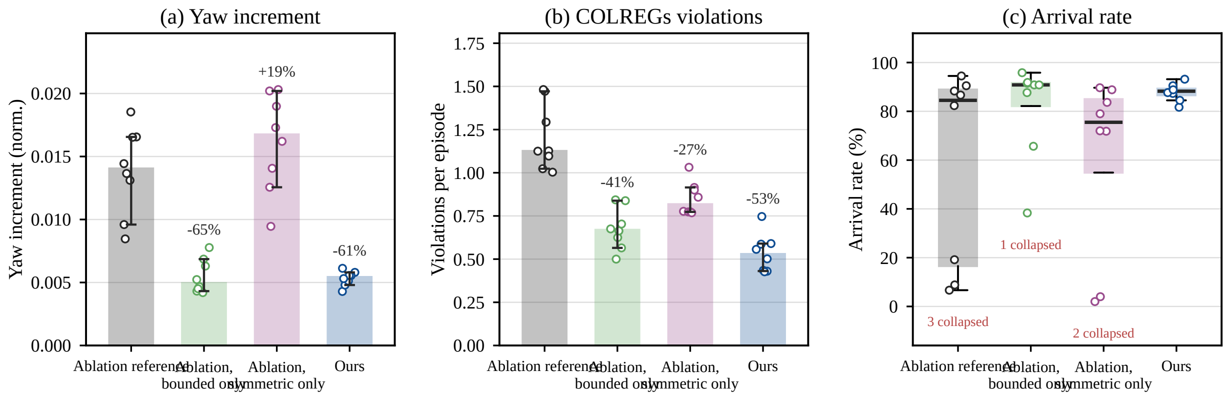

Figure 2 The 2×2 **ablation.** Paired by seed. Bars are medians across seeds, error bars are bootstrap 95% intervals, and open points are single seeds. The label above each bar is the change relative to the leftmost bar. The paired tests are reported in the text. The arrival rate is bimodal, so it is drawn as a box plot and the number of diverged seeds is printed below each box.

Second, reliability differs a great deal among the shielded configurations. Under the convergence rule declared in advance, all 8 seeds of Ours reach a working policy, and it is the only shielded configuration for which this holds. The shielded ablation configurations reach 5 to 7 seeds, and the discrete safe baseline reaches 6 (Figure 3e). The two changes therefore improve more than the final numbers. They also lower the chance that training itself fails. Diverged seeds are counted as they are, and a converged-seeds-only version is given.

A counterexample that has to be stated. On the test set of 600, 23 of the 48 shielded runs have a collision rate of exactly zero, and 25 do not. The largest is 1.50%, on two seeds of the discrete safe baseline. For Ours, 6 of the 8 seeds have no collision at all and the other two have exactly one collision each, which is 0.17%. The earlier figure of zero came from the validation set of 100 scenarios, where a single collision is below the resolution of the sample. This agrees with the scope statements of Section 4. The guarantees by construction cover one-step collision freedom in the far field and the turn direction on give-way steps. They do not cover emergency situations or the fallback. A collision rate of zero is therefore an observation and not a guarantee.

5.6 Behavior of the Safety Shield

The emergency branch is the only part of this method that offers no guarantee, so how often it is used and what follows from its use are reported separately. At the end of training, the projection produced 93.6% of the control commands. The emergency controller produced 6.2%, and the three fallback branches produced 0.22% in total (Figure 4a). On the test set the share of emergency steps is 2.42%, against 4.64% for the discrete safe baseline. The fallback is rare because it is used only when the safe control set is empty, and that seldom happens outside an emergency situation. Collisions are rare enough that their source cannot be attributed with confidence. Ours records two collisions in 4,800 test episodes. If a collision falls under emergency control, then it occurs on a step where the last resort had already been engaged and had failed. Those steps lie outside the scope of Proposition 1 by definition, since that proposition covers the far field, non-emergency, single-step case. This is not a counterexample to it, but it does confirm that the emergency branch is an empirical fallback.

The size of the projection correction is normalized by the half-width of the action box. At the end of training its median is 0.082 and its interquartile range is [0.065, 0.102]. The gap between the command of the policy and what the rules require therefore stays within about a tenth of the control range. It also narrows during training (Figure 4c). Among the four shielded continuous configurations the correction is largest for the symmetric-only configuration, at 0.093, and smallest for the bounded-only configuration, at 0.066. That ordering matches the turn increments in Figure 2.

Online cost. The projection is one quadratic program with two variables and at most six constraints, so it is cheap enough to run at every step. The timings below come from a single thread of the 32-core Linux server that ran the evaluation ([TBD]), over 1,872 decision steps of the trained policy. The median solve time of the program is 4.11 ms and the 99th percentile is 8.85 ms. The forward pass of the policy adds a median of 0.44 ms. A whole decision step, including the environment, has a median of 5.30 ms and a 99th percentile of 10.14 ms. The decision period is 10 s,

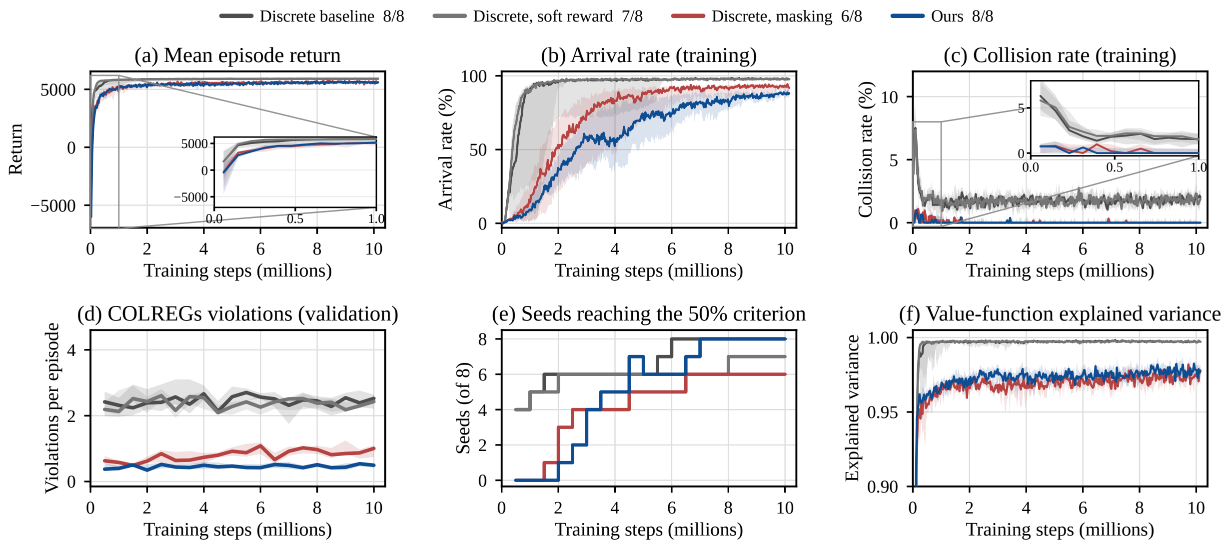

Figure 3 Training curves. Thick lines are medians and shaded bands are interquartile ranges over **converged seeds only**, because a diverged seed pulls the band to zero. The fraction in the legend is the number of converged seeds out of eight. Panels (a) to (c) and (f) are recorded once per rollout on the training scenarios, so they carry exploration noise. Panel (d) is measured on the validation set once per checkpoint, which is why it has fewer points. Panel (e) counts all eight seeds. **The reward function differs across configurations, so the level in panel (a) is not compared across them.**

so the 99th percentile of a whole step is about one thousandth of the period. **These are wall-clock times on one machine. They record an order of magnitude. They are not a comparison against any other method, which would need the same machine and the same protocol.**

5.7 Comparison with External Baselines

Three geometric collision-avoidance controllers that need no training are run through the same evaluation pipeline, on the same test set, at the same sample size (Table 4). The question here is not whether a learning method is stronger. It is which dimensions a geometric method already covers.

The answer narrows the defensible claim to one dimension. The collision rate does not separate the two families, since the control barrier function filter also reaches 0.00% on this test set. The arrival rate and the smoothness columns are governed by the type of terminal constraint and by the control range, and neither supports a reading in favor of Ours. The real difference is in COLREGs violations, and almost all of it comes from stand-on violations. These controllers aim only at avoiding contact, so they keep maneuvering during periods in which the rules require course and speed to be held. Violations are decided by the offline scorer from the bare situation predicates, so the comparison is made under one condition. The claim of this paper against geometric methods rests on this dimension alone. Ours records 0.53 violations per episode against 4.95 for the control barrier function filter at its best setting, which is a factor of 9.3. Almost all of that gap is in stand-on violations, at 0.36 against 4.48.

6 DISCUSSION

6.1 Three Points about the Mechanism

A soft reward cannot exclude a command. A soft reward can only shift the expected behavior of the policy. Masking and projection can remove a command outright. The difference between them is the premise. Masking needs the safe control set to be enumerable. This paper replaces that premise by convexity. The compliant direction is exactly one half-plane, and projecting onto its intersection with the action box is a quadratic program in two variables.

Both refinements belong to the continuous action space. A quantized action space has no samples pushed past the action box, and it has no channel whose differential entropy is unbounded above. These two problems appear only after a COLREGs shield is moved into a continuous action space. The ablation is therefore not the tuning of two

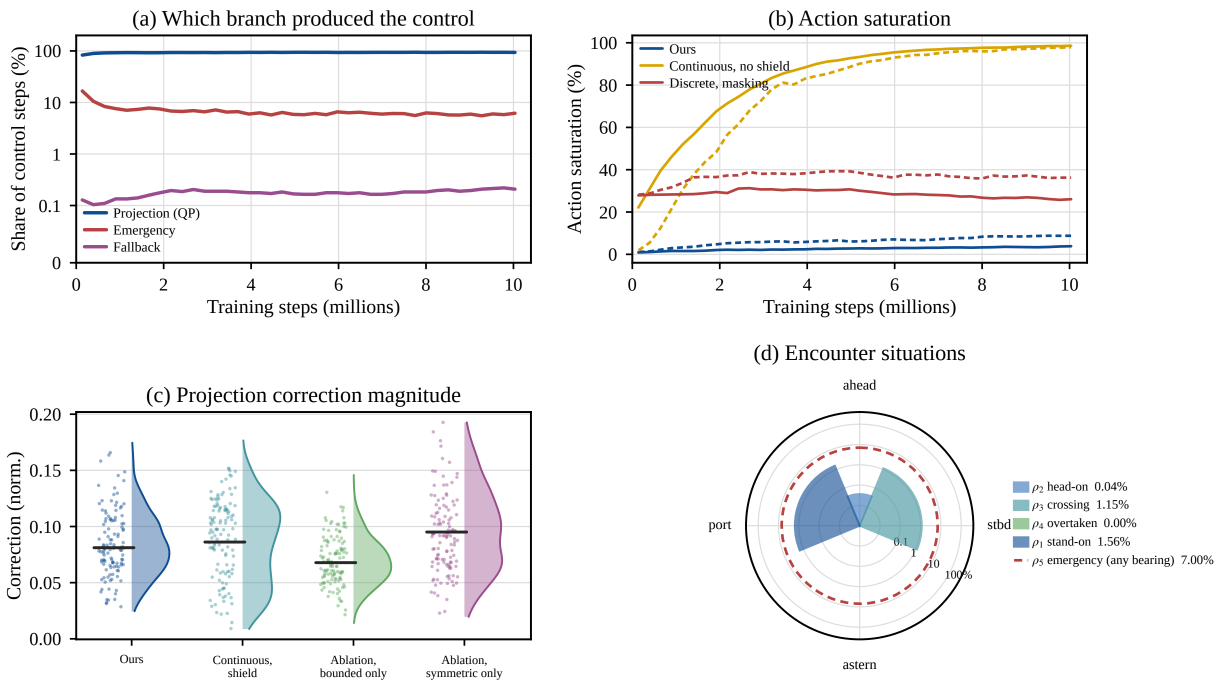

Figure 4 Behavior of the shield. In panel (d) the angle is the sector in which the convention defines each situation, **not** the measured bearing distribution of target vessels, and the radius is logarithmic. This figure is collected during training rather than at evaluation.

TABLE 4 External geometric baselines. Official test set, $N = 600$.

Method	Control range	Arrival %	Coll. %	COLREGs violations / episode			Turn Δ	Jerk
				Total	Give-way	Stand-on		
Control barrier function	Policy range	70.17	0.00	4.95	0.47	4.48	0.00242	0.171
	Policy range, margin	69.83	0.00	4.49	0.45	4.04	0.00235	0.167
	Actuator limits	69.33	0.00	6.48	0.52	5.97	0.00058	0.050
PD controller	Policy range	58.83	18.00	9.79	0.44	9.35	0.00492	0.310
	Actuator limits	57.67	16.67	8.10	0.63	7.47	0.00032	0.040
Velocity obstacle	Policy range, margin	57.00	0.00	8.20	0.50	7.70	0.00425	0.279
	Actuator limits	42.67	0.00	8.06	0.68	7.38	0.00128	0.071
Ours	Policy range			(see Table 3)				

Notes. These controllers need no training and have no seed. Each is reported at its own best setting, tuned on a set that is disjoint from the one used here. The policy range is ± 0.048 and ± 0.018, the same authority as the trained policies, and the actuator limits are ± 0.24 and ± 0.03. Both are reported so that a difference in authority is not a confounder. **The arrival rate is not a measure of collision avoidance.** These controllers have no terminal alignment stage. They reach the goal position in 96.8% to 99.7% of episodes and then fail the terminal heading gate.

hyperparameters. It repairs two defects created by that move.

The mechanism complements the control-theoretic shields. A reactive barrier function shield has no recursive feasibility proof under a 10 s zero-order hold. The clearable set of this work suggests one route to that property, but the closed-loop integration and the recomputation under the official evaluation pipeline are not finished. It is stated here as a design direction and not as a guarantee that has been met.

6.2 What Projection Buys and What It Costs

Both sides of the exchange can be measured directly, and Figure 5 reports them.

The gain is in the resolution of a compliant turn. The rules require the heading change over the maneuver window to reach a threshold, which converts to a smallest admissible turn rate ω_{turn}. A continuous command reaches that value exactly. A finite grid can only take the nearest grid point at or beyond it, so it always overshoots. On the 7×7 grid reproduced here, the smallest available compliant turn rate is 37.5% above that value.

That gap does not shrink monotonically as the grid is refined. The overshoot is set by the divisibility of ω_{turn} by the grid spacing, not by the spacing itself. In Figure 5a a 5×5 grid overshoots by 3.1% while a 7×7 grid overshoots by 37.5%. The common statement that a finer grid approaches continuous control therefore does not hold for this task. Only a continuous command with projection reaches the smallest admissible turn rate.

The cost is in computation, and it runs against intuition. Figure 5b shows that enumerating and masking $K \times K$ commands grows with K^2. Even at $K = 121$, which is 14,641 commands, it stays well below the time of one projection. **Projection is not the cheaper option.** Its justification is capability rather than speed. Enumeration has no meaning on a continuous set, while projection is feasible and lands exactly on the boundary of the constraint. The deployment cost of one step is reported separately in Section 5.6.

The difference in throttle resolution, where the discrete configurations have only seven levels, is structural in the same way. It cannot be read directly from the throttle increment column, because that column is also affected by the action-increment penalty, and that penalty is active only in the continuous configurations. The attributable comparison holds only among configurations that run the same penalty.

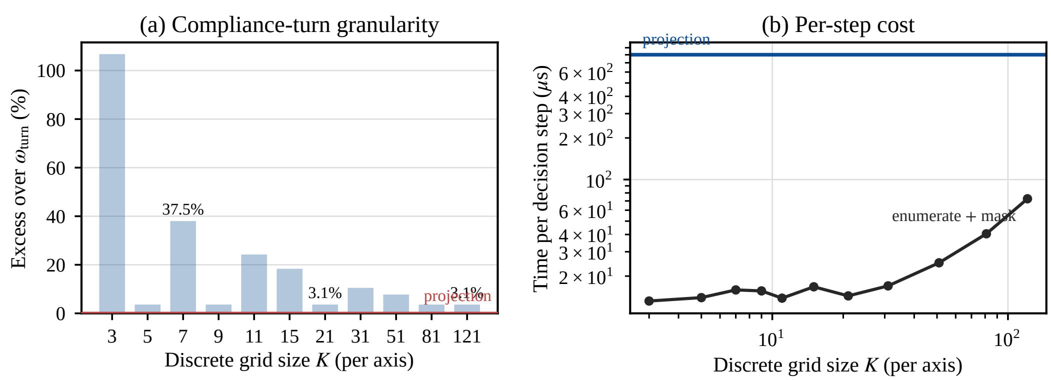

Figure 5 Projection against enumeration. (a) The amount by which the smallest compliant turn rate available on a $K \times K$ grid exceeds the smallest admissible value ω_{turn}. It is set by divisibility and **does not fall monotonically with** K. A continuous command with projection reaches it exactly, shown by the red line. (b) Time for one decision step. Enumeration grows with K^2, while projection does not depend on K. Panel (a) is exact arithmetic. Panel (b) is measured on one machine and is used only for the growth in K, not as the deployment cost.

6.3 What This Means for Putting Autonomous Vessels into Service

Making compliance a projectable hard constraint changes more than the behavior of the policy. It changes how this layer can be reviewed. When compliance is induced by training, verification can only be statistical. A sample of scenarios is run, a violation frequency is reported, and the conclusion moves with every version of the policy. When compliance holds by construction, the object of verification becomes the constraint itself. A reviewer checks whether the state machine classifies the situation correctly, whether the half-plane points the way the clause requires, and whether the projection lands inside the feasible set. All three can be checked step by step, and none of them depends on the training outcome. For type approval and for accident review this is the more workable form. The reviewer does not need to reproduce the training run.

The mechanism also does not require the existing control system to be replaced. It accepts a desired control command from any source and returns the nearest compliant command. It can therefore sit as a separate compliance layer above an existing autopilot or path-following controller. Its online cost is one quadratic program per step, with two variables and at most six constraints, and it needs no dedicated hardware. Deployment on current shipboard computing platforms is realistic.

7 CONCLUSIONS

The give-way clauses of COLREGs require a vessel to turn to a stated side, so compliance cannot be verified from whether contact occurred. Existing work either buys a provable form of compliance with a discrete action space and masking, or buys control resolution with continuous commands and a soft reward. This paper shows that the exchange is not necessary. The give-way clause is exactly a half-plane in the control plane. The premise that the safe set must be enumerated can therefore be replaced by the premise that it is convex. A projection-based safety shield is therefore brought to maritime COLREGs under continuous control. Directional compliance on every give-way step where the projection is feasible holds by construction, independently of the training outcome and of the seed. Any behavioral rule that can be written as a convex constraint in the action space can follow the same route.

On the official test set of 600 scenarios the median violation count of Ours is 0.53 per episode. The discrete shielded baseline records 1.16 and the strongest geometric baseline records 4.95. It is the only shielded configuration that meets the convergence criterion on all eight seeds.

The boundaries of these conclusions are as follows. The shield does not guarantee freedom from collision. Directional compliance holds on **every give-way step where the projection is feasible**, not on every step. An emergency situation bypasses the constraint by design, so the measured violation count is not zero. Propositions 1 and 3 both assume that the target vessel holds its velocity, and a maneuvering target vessel voids them. Neither the arrival rate nor smoothness is claimed as a benefit, and smoothness cannot be attributed across action spaces because the action-increment penalty is active only in the continuous configurations. Two biases in the reported numbers are known, namely checkpoint selection and machine-to-machine variation. The main table is therefore reported in two versions, and every configuration in one table is evaluated on one machine in one pass. On rule coverage, the directional requirements of Rules 13 to 17 and the stand-on obligation to hold course and speed are implemented. The stand-on duty to take avoiding action on its own is not. The overtaking branch is implemented but the scenario library contains no overtaking case, so it is untested. The discrete baseline is a reimplementation, and its seed fragility is attributed to that reimplementation.

Three lines of work follow. The terminal constraint of the backup maneuver should be wired into the deployed shield and verified in closed loop. Directional compliance should be extended to several target vessels with a conflict resolution rule. The far-field test should be implemented as an $O(1)$ early exit.

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